

Graphics Card Fundamentals

What are graphics cards? Why do we need these? How do they work? Find out



ATI Radeon HD 4770 graphics card (Courtesy: Wikipedia)

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Graphics cards are a very important hardware for displaying 3D and 2D scenes, especially in gaming, with minimum flickering. These are therefore also called display cards or video cards.

A graphics card takes the binary input from a CPU operated with a software and converts this binary data into a picture where the decision about each pixel of the display is done precisely while adding sufficient lighting, texture and colour. It can also be referred to as an expansion card since it provides a series of output images.

Fig. 1 shows how a graphics card looks like. Sometimes the motherboard too includes a graphics chipset. This chipset uses part of the system's RAM and is known as onboard or integrated graphics chipset. However, its performance is not up to the mark, rendering it undesirable for 3D applications.

For production of high-texture 3D and 2D scenes, such as in fast-paced games, complex mathematical and geometrical calculations are to be done at an extremely fast pace in order to eliminate the flickering problem. But the CPU of a computer is not designed for executing such a single bulk operation since it has to perform other essential operations too. Hence an external hardware is required for performing

this operation and so we use a graphics card.

A graphics card has its own RAM and processor. The RAM is dual ported and hence the processor can read from it and write into it simultaneously.

All the motherboards allow disabling of the integrated graphics chip in BIOS and provide a peripheral component interconnect (PCI), advanced graphics port (AGP) or PCI-Express (PCIe) slot for connecting a high-performance graphics card.

The processor of the graphics card is also known as graphics processing unit (GPU) and is specifically designed for performing all the complex mathematical operations. The design uses an architecture different from a normal processor due to its complex operations to increase the efficiency and speed and to deliver a high performance. But it produces lots of heat and so we find heat-sink surrounding it and a fan above it to avoid overheating.

It has a BIOS chip, which contains the settings, i.e., a basic program that governs the video card's operations and allows the software to interact with the card and even perform error check during startup.

The settings in BIOS can be changed for overclocking, which poses a high risk to the card leading to irreversible damage. The GPU uses a special program to process the data.

To improve the quality of scenes, full-scene anti-aliasing and anisotropic filtering are employed by the processor. Full-scene anti-aliasing helps in providing a pleasing image by smoothening the edges of objects by removing the jagged pixels. Anisotropic filtering is said to maintain crisp texture detail at all viewing orienta-



Fig. 1: A graphics card